Name bhavya rana
Sapid 590034689
Batch 15

Algorithm

1. Start.
2. Read the number of disks `n`.
3. Call `towerOfHanoi(n, A, B, C)`, where:
   - `A` = Source rod
   - `B` = Auxiliary rod
   - `C` = Destination rod
4. If `n == 1`, move disk 1 directly from the source to the destination.
5. Otherwise:
   - Recursively move `n-1` disks from source to auxiliary.
   - Move disk `n` from source to destination.
   - Recursively move `n-1` disks from auxiliary to destination.
6. Stop.

Test Case Table

Test Case	Number of Disks `n`	Number of Moves
1	1	1
2	2	3
3	3	7
4	4	15
5	5	31

The minimum number of moves required is:

$$\boxed{2^n - 1}$$

C program

```
#include <stdio.h>
```

```c
void towerOfHanoi(int n, char source, char auxiliary, char destination)
{
    if (n == 1)
    {
        printf("Move disk 1 from %c to %c\n", source, destination);
        return;
    }

    // Move n-1 disks from source to auxiliary
    towerOfHanoi(n - 1, source, destination, auxiliary);

    // Move largest disk from source to destination
    printf("Move disk %d from %c to %c\n", n, source, destination);

    // Move n-1 disks from auxiliary to destination
    towerOfHanoi(n - 1, auxiliary, source, destination);
}

int main()
{
    int n;

    printf("Enter number of disks: ");
    scanf("%d", &n);

    towerOfHanoi(n, 'A', 'B', 'C');

    return 0;
}
```

Execution

